



Application of the SeDeM system for the preparation of antiparasitic tablets from mesquite flour for use in sheep

Aline Cristina Cazares-Solis¹ · Verónica Rodríguez-Guerrero¹ · Oswaldo Castañeda Hernández² · Luz María Melgoza Contreras¹

Received: 10 October 2023 / Accepted: 6 February 2024 / Published online: 17 February 2024
© The Author(s), under exclusive licence to Springer Nature B.V. 2024

Abstract

The use of herbal medicine to treat various diseases is becoming increasingly important as an alternative therapy. Numerous plants have been traditionally used for different purposes, including antiparasitic in humans and animals. Diseases caused by gastrointestinal parasites in ruminants, especially by the nematode *Haemonchus contortus*, cause large economic losses to the producers, whether by complications of the diseases or the cost of treatment. The main way of handling nematodiasis is by administering anthelmintic drugs, but their excessive use has the disadvantage of causing drug resistance; therefore, an alternative is the use of herbal medicine for this purpose. Mesquite (*Prosopis* spp.) has been used in Mexico to treat gastrointestinal diseases attributed to helminths. The present study aimed to characterize the rheological properties of mesquite flour using the SeDeM Expert System to determine its suitability for tablet production by direct compression. Direct compression technology facilitates the tableting process by reducing manufacturing costs. The results of the present study indicate that mesquite flour can be processed by direct compression. The latter could allow the manufacturing of economic tablets to treat infections by *H. contortus* in ruminants.

Keywords Ruminants · Anthelmintic plants · Mesquite flour · Tablets by direct compression · SeDeM Expert System

Introduction

Mexico's great plant diversity and cultural richness have benefited the use of plants for medicinal purposes since the pre-Hispanic ages. An extensive variety of phytotherapeutic treatments are part of traditional Mexican herbal medicine. These plants are used as nutraceuticals and are becoming increasingly relevant as alternative therapies to treat some diseases. Among these plants are those used as antiparasitic and anthelmintic (Moya and Escudero 2015).

Parasitic diseases are one of the main issues affecting human and animal health. In the veterinary field, gastrointestinal parasites affect mostly ovines, goats, and bovines. One of the most important parasites is *Haemonchus contortus*,

which causes malnutrition, anemia, stunted growth, loss of appetite, decreased weight gain, or even death, leading to economic losses for cattlemen (Muñoz-Guzmán et al. 2016; Pinilla et al. 2018).

Nowadays, this nematode is primarily controlled by administering anthelmintics. Although this is efficient, it poses certain disadvantages, such as drug resistance, risk of products and environmental contamination, and the cost of treatment (Munguía et al. 2013; Besier et al. 2016; Reyes-Guerrero et al. 2021). Hence, alternative strategies have been adopted against haemonchosis, such as using herbal products like mesquite as a phytotherapeutic agent and for its nutritional value for ruminants (nutraceutical).

Mesquite (*Prosopis* spp.) is a legume distributed in Mexican arid, semiarid, and warm weather regions (Andrade-Montemayor et al. 2011). Mesquite has been used, by indigenous people, as food and treatment for gastrointestinal diseases (Reyes et al. 2017; Reyes-Guerrero et al. 2021). It has anthelmintic potential as a natural and inexpensive alternative to control nematodiasis in cattle (López-Arroche et al. 2008; De Jesús-Gabino et al. 2009). Tablets could be manufactured using mesquite flour for

✉ Luz María Melgoza Contreras
lmelgoza@correo.xoc.uam.mx

¹ Departamento de Sistemas Biológicos, Universidad Autónoma Metropolitana-Xochimilco, Mexico City, Mexico

² Doctorado en Ciencias Biológicas y de La Salud, Universidad Autónoma Metropolitana-Xochimilco, Mexico City, Mexico

easier administration and at necessary dosages to treat haemonchosis.

Tablets are a solid pharmaceutical dosage form obtained from the mechanical compression of granulates or a mixture of powders from one or various active pharmaceutical ingredients (APIs) with the addition of different excipients (García and Santos 2001). Products made by direct compression are inexpensive due to the few operational units required. The SeDeM Expert System is a methodology applied in tablet preformulation studies; it determines the capacity of a powder to be processed by direct compression. It measures 12 parameters related to the rheological behavior of APIs or excipients (Table 1).

The results obtained experimentally are converted to a radial scale (0 to 10) and then presented in a 12-sided regular polygon. This methodology allows determining numerically the suitability of a powder by calculating three indices: parametric index (PI), parameter profile index (PPI), and good compressibility index (GCI) (Suñe-Negre et al. 2014). The results can be used to characterize the physical properties of the powder, to indicate if it is suitable for direct compression or to identify its deficiencies and to know the quantities to be used to improve those deficiencies. Reports on the application of the SeDeM Expert System to herbal products are scarce (Wan et al. 2019; Zhang et al. 2021); hence, in the present study, mesquite flour is characterized using the SeDeM Expert System to propose the manufacturing of mesquite flour tablets by direct compression as a treatment for the *H. contortus* infection in sheep.

Materials and methods

Materials

Mesquite pods

For the application of the SeDeM Expert System and preparation of tablets was used mesquite flour.

Mesquite pod picking

It was carried out in the arid ecoregion of Hidalgo, Mexico, during July, where the average annual temperature is 16 °C and the average rainfall is 800 mm yearly (INEGI, s.f.). The pods were collected by hand directly from the trees and randomly, considering the ripe pods according to Cerón-Ortiz et al. (2015), and that showed no visible physical damage before they fell to the ground. In the end, the pods that did not present deterioration due to attack by insects or fungi and had a similar morphology were selected.

Methods

Dried and ground of mesquite pods

The samples were dried in an oven (Felisa, FE-293) at 55 °C for 72 h, then ground in a Thomas-Wiley Model 4 Mill (Thomas Scientific, USA) in a 1-mm mesh; the mesquite flour obtained was stored in hermetic plastic bags, and they were identified for further analysis.

Table 1 Parameters used in SeDeM Expert System

Incidence	Parameter	Symbol	Unit	Acceptance limits	Equation for normalization
Dimensions	Bulk density	Db	g/mL	0–1	10v
	Tapped density	Dt	g/mL	0–1	10v
Compressibility	Interparticle porosity	Ie	-	0–1.2	10v/1.2
	Carr index	CI	%	0–50	v/5
	Cohesion index	Icd	N	0–200	v/20
Flowability/powder flow	Hausner index	HI	-	1–3	(30 – 10v)/2
	Angle of repose	(α)	°	0–50	10 – (v/5)
	Powder flow	t"	s	0–20	10 – (v/2)
Lubricity/stability	Loss on drying	LOD	%	0–10	10 – v
	Hygroscopicity	%H	%	0–20	10 – (v/2)
Lubricity/dosage	Particles < 50 μm (fine)	%Pf	%	0–50	10 – (v/5)
	Homogeneity index	IΘ	-	0–2 × 10 ⁻²	500v

v experimental value

SeDeM system characterization of mesquite flour

The SeDeM parameters were calculated by triplicate, and the results represent the average of measurements.

Bulk and tapped density (Db and Dt)

Bulk density was determined according to method 1 of the MGA 1031 of the FEUM (2018). A mesquite flour sample of 100 g was poured into a 250-mL graduated cylinder, and the volume was measured without compacting. The bulk density was calculated in grams per milliliter (g/mL).

To determine the tapped density, the same sample of bulk density was used without taking it out of the graduated cylinder. This was tapped manually 250 times on a smooth flat surface at a constant rate. Afterward, the compacted volume reading was taken, and the tapped density was calculated in grams per milliliter (g/mL) (FEUM 2018).

Interparticle porosity (Ie)

Interparticle porosity was calculated using the densities of the material, according to Eq. 1 (Sipos et al. 2017):

$$Ie = \frac{Dt - Db}{(Dt)(Db)} \quad (1)$$

Carr index (CI)

The Carr index was calculated using the densities of the material, according to Eq. 2 (Sipos et al. 2017):

$$CI = \frac{Dt - Db}{Dt} \times 100 \quad (2)$$

Cohesion index (Icd)

The cohesion index was determined by compacting a mixture of mesquite flour, talc (2.36%), AEROSIL® 200 (0.14%), and magnesium stearate (1.00%) in a Riva Piccola B-10 rotatory tablet press machine at the feasible maximal compaction pressure without producing lamination or capping. Convex tablets weighing 800 mg and 13 mm in diameter were obtained. The hardness was determined for the final product in an Erweka TBH 125 in Newton (Suñe et al. 2013).

Hausner index (HI)

The Hausner index was calculated using the densities of the material, according to Eq. 3 (Sipos et al. 2017):

$$HI = \frac{Dt}{Db} \quad (3)$$

Angle of repose (α)

The angle of repose was determined according to the MGA 1061 of the FEUM (2018). A steel funnel with the inferior hole blocked was placed on a support 10 cm from the table's surface. Subsequently, a 100 g sample was poured into the funnel and made to flow through the funnel. The diameter and height of the cone formed by the powder were determined to obtain the angle of repose (Eq. 4).

$$\tan \alpha = \frac{h}{r} \quad (4)$$

where h is the height of the cone and r is the radius.

Powder flow (t'')

The powder flow was measured in the same way as the angle of repose, using a 100 g sample. The time taken for 100 g of the sample to flow through the funnel was determined in seconds (FEUM 2018).

Loss on drying (LOD)

Loss on drying was determined using a 1 g sample in a thermobalance ROCA®, according to Eq. 5.

$$LOD = \frac{W_f}{W_i} \times 100 \quad (5)$$

where W_i is the initial weight and W_f is the final weight.

Hygroscopicity (%H)

Hygroscopicity was determined by placing an accurately weighed sample in a desiccator and saturated sodium chloride solution (40 g NaCl in 100 mL water). Humidity was allowed to equilibrate in the chamber for 24 h before placing the samples. Subsequently, three samples were placed in the desiccator at $76 \pm 2\%$ relative humidity and $22 \pm 2^\circ\text{C}$ for 24 h. The hygroscopicity was calculated as Eq. 6 (Salim et al. 2021).

$$\%H = \frac{W_f - W_i}{W_i} \times 100 \quad (6)$$

where W_i is the initial weight and W_f is the final weight.

Homogeneity index (I_θ)

A laser diffraction particle size analyzer LA-950V2 HORIBA was used with the dry dispersion module to determine the particle size. The results were divided into fractions < 50, 50–100, 100–212, 212–355, and 355–500 μm. The homogeneity index was calculated from the experimental results, according to Eq. 7 (Suñe et al. 2013):

$$I_{\theta} = \frac{F_m}{100 + (d_m - d_{m-1})F_{m-1} + (d_{m+1} - d_m)F_{m+1} + (d_m - d_{m-2})F_{m-2} + (d_{m+2} - d_m)F_{m+2} + \dots (d_m - d_{m-n})F_{m-n} + (d_{m+n} - d_m)F_{m+n}} \quad (7)$$

where.

- F_m Percentage of particles in the majority range.
- F_{m-1} Percentage of particles in the range immediately below the majority range.
- F_{m+1} Percentage of particles in the range immediately above the majority range.
- n Order number of the fraction studied under a series, regarding the majority fraction.
- d_m Mean diameter of the particle in the majority fraction.
- d_{m-1} Mean diameter of the particles in the fraction of the range immediately below the majority range.
- d_{m+1} Mean diameter of the particle in the fraction of the range immediately above the majority range.

Percentage of particles < 50 μm (% Pf)

The percentage of the particles measuring < 50 μm was determined along with the homogeneity index test and represents the fine particles.

Data processing

Once the SeDeM parameters were established, the radius was calculated, according to the factors applied shown in Table 1.

Calculation of acceptability indexes

From the parameters described above, the parametric index, the parametric profile index, and the good compression index were calculated using the following equations (Suñe et al. 2013):

Parametric index (PI) Number of parameters whose value

is equal to or greater than 5 ($n^{\circ}P \geq 5$) divided by the total number of parameters studied ($n^{\circ}Pt$) (Eq. 8).

$$PI = \frac{n^{\circ}P \geq 5}{n^{\circ}P_t} \quad (8)$$

The minimum expected value for a good substance is $PI \geq 0.5$.

Parameter profile index (PPI) Arithmetic mean of the radius value (r) of all parameters studied (Eq. 9).

$$PPI = \frac{\sum_{i=1}^n r_i}{n} \quad (9)$$

where $\sum_{i=1}^n r_i$: the sum of the radius values of all parameters studied.

n : the number of total parameters studied.

The minimum expected value for a powder of suitable characteristics is $PPI \geq 5$.

Good compressibility index (Eq. 10) (GCI)

$$GCI = PPI \times f \quad (10)$$

where PPI: the parametric profile index.

f is a reliability factor that, when performing the determinations for the 12 parameters set in the SeDeM system, has a value of 0.952.

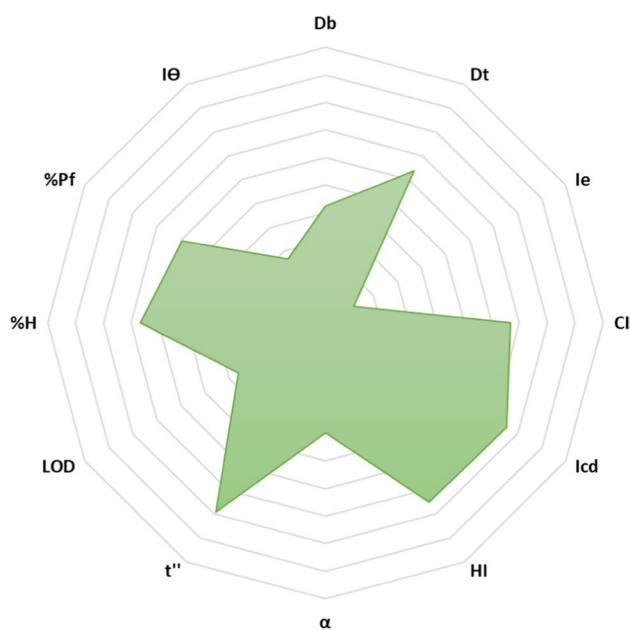
The minimum acceptable value to consider a powder as acceptable to be used in direct compression processes is $GCI \geq 5$.

Results

The values given in Table 2 are the average value of three measurements. The values obtained are shown for each test as the radius and the incidences obtained after data processing. The parameters in which mesquite flour exhibits

Table 2 Parameters, average of incidence, and acceptability indexes of mesquite flour

Incidence	Parameter	Experimental value	Radio	Average of incidence
Dimensions	Db (g/mL)	0.42	4.24	5.31
	Dt (g/mL)	0.64	6.38	
Compressibility	Ie	0.14	1.18	5.14
	CI	33.42	6.68	
	Icd (N)	151.06	7.55	
Flowability/powder flow	HI	1.50	7.49	6.45
	α (°)	30.14	3.97	
	t''(s)	4.19	7.91	
Lubricity/stability	LOD (%)	4.96	3.60	5.13
	H (%)	6.66	6.67	
Lubricity/dosage	Pf (%)	20.21	5.96	4.32
	I θ	0.0054	2.68	
Parametric index			0.58	
Parameter profile index			5.36	
Good compressibility index			5.10	

**Fig. 1** SeDeM diagram of mesquite flour

deficiencies are bulk density, interparticle porosity, angle of repose, loss on drying, and homogeneity index. In all these parameters, the radius obtained was below the minimum acceptable value (MAV) of 5 (Salim et al. 2021). The radius values are presented graphically in Fig. 1, revealing the deficiencies. The parameters with scarce shaded areas portray flour deficiencies.

The averages of the incidences show deficiencies in lubricity/dosage, obtaining value of 4.32, below the MAV

of 5. The deficiencies in lubricity/dosage involve rheological characteristics concerning the shape and the particle size distribution, which can be corroborated by observing the radius obtained for the homogeneity index (2.68).

Although the interparticle porosity index had a smaller radius (1.18) than all parameters, the compressibility incidence value (5.14) was above the MAV, because of binder property of the mesquite flour (Suñe et al. 2013).

The results of the acceptability index are shown in Table 2. Mesquite flour exhibits an acceptable PI of 0.58, indicating that 7 of the 12 parameters obtained a radius above the MAV, just as PPI and GCI presented values above the MAV 5.36 and 5.10, respectively. Thus, mesquite flour is suitable for direct compression (Scholtz et al. 2017).

Discussion

According to the SeDeM Expert System the PPI and GCI parameters (5.36 and 5.10, respectively), mesquite flour may perform well to produce tablets through direct compression material (Suñe et al. 2013). But to improve the performance in lubricity/dosage of the mesquite flour in direct compression, excipients addition suggested. Excipients like microcrystalline cellulose, fine particle ethyl cellulose, or spray-dried lactose are used to correct the compressibility and lubricity/dosage issues. It is known that microcrystalline cellulose shows a plastic deformation which helps when producing tablets with good compactibility (Rojas et al. 2013). Fine particle grade ethyl cellulose has good compression properties since its particle size increases interparticle bonding for more contact points (Ghori and Conway 2016). Spray-dried lactose enhances flowability of powders for its homogeneity particle size and spherical shape and also improves the compactibility because of your plastic deformation (Al-Zoubi et al. 2021). Another way to prove the mesquite flour performance is by drying (50–65 °C) before handling. The humidity percentage of the flour (4.96%), which is common in mesquite flours (Díaz-Batalla et al. 2018; Gonzales-Barrón et al. 2020; De Melo-Cavalcante et al. 2022), exceeds the optimal value proposed by the SeDeM methodology (1–3%). The high percentage of humidity on mesquite flour decreases its rheological characteristics, such as flow; it promotes caking because the water layer that coats the particles increases cohesion by forming liquid bridges. These changes in flowability affect the manufacture of tablets. Poor flowability hinders the flow of powder from the hopper to the matrix causing poor filling and, consequently, variation in dosage (Sandler et al. 2010; Crouter and Briens 2013).

Currently, only two studies evaluated the suitability of herbal extracts by direct compression were analyzed using the SeDeM Expert System. Wan et al. (2019) analyzed a

Rhodiola extract and the mixture of Rhodiola with excipients. On the other hand, Zhang et al. (2021) evaluated a co-processed *Zingiberis Rhizoma* with nano-silica; however, no study was found that analyzed mesquite flour.

In this study, mesquite flour was characterized for the first time using the SeDeM Expert System, identifying the values of PI, PPI, and GCI that indicate it can be compressed by direct compression technology.

It was made one lot of mesquite tablets weighing 800 mg and 13 mm in diameter (Dose 696 mg) by direct compression with microcrystalline cellulose. Avicel pH 102 (10%) (kindly donated by FMC, Mexico, batch: P207818346), colloidal silica dioxide 2.5% (HDK® N20, Wacker Chemie, AG, Munich, Germany; batch: YK47349), and magnesium stearate 0.5% (LIGAMED® MF-2 K, Peter Greven, Bad Muenstereifel, Germany; batch: C738499) were used as glidant and lubricant.

The tablets obtained weighed 785 ± 8.63 mg showing excellent mechanical properties, hardness (6.86 ± 1.20 Kp), friability (weight loss 0.06%), and a disintegration time between 23 and 37 min, which will allow the mesquite to be slowly released, and as its size decreases, it will facilitate the passage to the abomasum of ruminants.

Preparing mesquite tablets by direct compression, in high-speed machines, is an economical proposal to treat nematodiasis by *H. contortus* in ruminants.

Acknowledgements The authors thank Dr. Juan Carlos Ángeles Hernández from the Autonomous University of the State of Hidalgo for providing the mesquite used in the development of this work, and Verónica Rodríguez thank CONAHCYT for the postdoctoral stay 2022 (1).

Author contribution Aline Cristina Cazares-Solis: methodology, validation, investigation, and writing—original draft.

Verónica Rodríguez-Guerrero: conceptualization, formal analysis, supervision, and writing—reviewing and editing.

Oswaldo Castañeda Hernández: formal analysis, supervision, and writing—reviewing and editing.

Luz María Melgoza Contreras: resources, supervision, project administration, and writing—reviewing and editing.

Data availability The datasets analyzed during the current study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors declare no competing interests.

References

- Al-Zoubi, N., Gharaibeh, S. F., Aljaberi, A., & Nikolakakis, I. (2021). Spray Drying for Direct Compression of Pharmaceuticals. *Processes*, 9(2), 267. <https://doi.org/10.3390/pr9020267>
- Andrade-Montemayor, H., Córdova-Torres, A., García-Gasca, T., Kawa J., 2011. Alternative foods for small ruminants in semiarid zones, the case of Mesquite (*Prosopis spp.*) and Nopal (*Opuntia spp.*). *Small Ruminant Research*. 98(1–3): 83–92. <https://doi.org/10.1016/j.smallrumres.2011.03.023>
- Besier, R.B., Kahn, L.P., Sargison, N.D., Van Wyk, J.A., 2016. Diagnosis, Treatment and Management of *Haemonchus contortus* in Small Ruminants. *Advances in Parasitology*. 93: 181–238. <https://doi.org/10.1016/bs.apar.2016.02.024>
- Cerón-Ortiz, A.N., Vidal-Gómez, A.Y. and Ángeles Monroy, M.A., (2015). Impacto del estado de madurez de la vaina de mezquite (*Prosopis laevigata*) en su composición fisicoquímica posterior a su transformación en polvo. *Revista Congreso Nacional de Ingeniería y Tecnologías para el Desarrollo Sustentable*. 1, pp. 1-7. http://conainte.itsoeh.edu.mx/revista/docs/VOL1_01-07.pdf
- Crouter, A. Briens, L., 2013. The Effect of Moisture on the Flowability of Pharmaceutical Excipients. *An Official Journal of the American Association of Pharmaceutical Scientists*, 15(1), 65–74. <https://doi.org/10.1208/s12249-013-0036-0>
- De Jesús-Gabino, A.F., Mendoza, P., Salinas-Sánchez, D.O., López-Arellano, M.E., Liébano-Hernández, E., Hernández-Velázquez, V.M., Valladares-Cisneros, G., 2009. Anthelmintic effects of *Prosopis laevigata*-hexanic extract against *Haemonchus contortus* in artificially infected gerbils (*Meriones unguiculatus*). *Journal of Helminthology*. 84(1): 71-75. <https://doi.org/10.1017/S0022149X09990332>
- De Melo-Cavalcante, A.M., De Melo, A.M., Da Silva, A.V.F., Da Silva-Neto, G.J., Barbi, R.C.T., Ikeda, M., Silva, G.B., Steel, C.J., Silva, O.S.D., 2022. Mesquite (*Prosopis juliflora*) grain flour: new ingredient with bioactive, nutritional, and physical-chemical properties for food applications. *Future Foods*. 5: 100114. <https://doi.org/10.1016/j.fufo.2022.100114>
- Díaz-Batalla, L., Hernández-Urbe, J.P., Román-Gutiérrez, A.D., Cariño-Cortés, R., Castro-Rosas, J., Téllez-Jurado, A., Gómez-Aldapa, C.A., 2018. Chemical and nutritional characterization of raw and thermal-treated flours of Mesquite (*Prosopis laevigata*) pods and their residual brans. *CyTA - Journal of Food*. 16(1): 444–451. <https://doi.org/10.1080/19476337.2017.1418433>
- Farmacopea de los Estados Unidos Mexicanos. (FEUM), 2018. Secretaría de Salud, Comisión permanente de la. *Farmacopea de los Estados Unidos Mexicanos*. 12 ed. México.
- García, S.M.J., Santos, B.D., 2001. Formas sólidas orales. En Vila-Jato (Ed.), *Tecnología Farmacéutica*. Vol. II. Editorial Síntesis. Madrid. 87-105 pp.
- Ghori, M. U. & Conway, B., (2016) “Powder Compaction: Compression Properties of Cellulose Ethers”, *British Journal of Pharmacy* 1(1). <https://doi.org/10.5920/bjpharm.2016.09>
- Gonzales-Barron, U., Dijkshoorn, R., Maloney, M., Finimundy, T., Calhelha, R.C., Pereira, C., Stojković, D., Soković, M., Ferreira, I.C.F.R., Barros, L., Cadavez, V., 2020. Nutritive and Bioactive Properties of Mesquite (*Prosopis pallida*) Flour and Its Technological Performance in Breadmaking. *Foods*. 9(5): 597. <https://doi.org/10.3390/foods9050597>
- Instituto Nacional de Estadística y Geografía e Informática. (INEGI). (s.f.). Cuéntame de México. Hidalgo. Recuperado 26 de Abril de 2023. <https://cuentame.inegi.org.mx/monografias/informacion/hgo/territorio/clima.aspx?tema=me&e=13>
- López-Aroche, U., Salinas-Sánchez, D.O., Mendoza De Gives, P., López-Arellano, M.E., Liébano-Hernández, E., Valladares-Cisneros, G., Arias-Ataide, D.M., Hernández-Velázquez, V., 2008. In vitro nematocidal effects of medicinal plants from the Sierra de Huautla, Biosphere Reserve, Morelos, Mexico against *Haemonchus contortus* infective larvae. *Journal of Helminthology*. 82(1): 25–31. <https://doi.org/10.1017/S0022149X07873627>
- Moya, M., Escudero, V., 2015. Las plantas medicinales en el control de nemátodos gastrointestinales en cabras: potencial de las plantas que crecen en la región de Coquimbo, Chile. *Revista Brasileira de Plantas Medicinai*. 17(3): 480-494. https://doi.org/10.1590/1983-084X/13_103

- Munguía-Xóchihua, J.A., Valenzuela-Medrano, J.C., Leyva-Corona, J.C., Morales-Pablos, M.I., Figueroa-Castillo, 2013. Potencial del orégano como alternativa natural para controlar *Haemonchus contortus* en ovinos de pelo. *Revista Latinoamericana de Recursos Naturales*. 9(1): 150–154. <https://revista.itson.edu.mx/index.php/rln/article/view/222>
- Muñoz-Guzmán, M.A., Sánchez-González, V.H., Revilla, V.A., Abd-Elghany, H.A., Cuenca-Verde, C., Cuéllar-Ordaz, J.A., Alba-Hurtado, F., 2016. Efecto de la hemoncosis experimental sobre las concentraciones séricas de sodio, potasio y cobre en dos razas ovinas. *Quehacer Científico en Chiapas*. 11(2): 68–75. https://www.dgip.unach.mx/images/pdf-REVISTA-QUEHACERCIENTIFICO/2016-jul-dic/Efecto_de_la_hemoncosis_experimental.pdf
- Pinilla, J.C., Flórez, P., Sierra, M., Morales, E., Sierra, R., Vásquez, M.C., Tobon, J.C., Sánchez, A., Ortiz D., 2018. Prevalencia del parasitismo gastrointestinal en bovinos del departamento Cesar Colombia. *Revista de Investigaciones Veterinarias del Perú*. 29(1): 278–287. <https://doi.org/10.15381/rivep.v29i1.14202>
- Reyes López, L. G. (2017). Caracterización de polvos (harinas) de mezquite (*Prosopis laevigata*) para aplicaciones alimentarias. Tesis de maestría (no publicada). Centro Interdisciplinario de Investigación para el Desarrollo Integral Regional Unidad Oaxaca, Instituto Politécnico Nacional. México.
- Reyes-Guerrero, D. E., Olmedo-Juárez, A., Mendoza-de Gives, P., 2021. Control y prevención de nematodosis en pequeños ruminantes: antecedentes, retos y perspectivas en México. *Revista Mexicana de Ciencias Pecuarias*. 12:186–204. <https://doi.org/10.22319/rmcp.v12s3.5840>
- Rojas, J., Aristizabal, J., & Henao, M. (2013). Screening of several excipients for direct compression of tablets: A new perspective based on functional properties. *DOAJ (DOAJ: Directory of Open Access Journals)*.
- Salim, I., Olowosulu, A.K., Abdulsamad, A., Gwarzo, M.S., Khalid, G.M., Ahmad, N.T., Eichie, F.E., Kurfi, F.S., 2021. Application of SeDeM Expert System in the development of novel directly compressible co-processed excipients via co-processing. *Future Journal of Pharmaceutical Sciences*. 7(135):1–12. <https://doi.org/10.1186/s43094-021-00253-z>
- Sandler, N., Reiche, K., Heinämäki, J., Yliruusi, J., 2010. Effect of Moisture on Powder Flow Properties of Theophylline. *Pharmaceutics*. 2(3): 275–290. <https://doi.org/10.3390/pharmaceutics2030275>
- Scholtz, J.C., Steenekamp, J.H., Hamman, J.H., Tiedt, L.R., 2017. The SeDeM Expert Diagram System: Its performance and predictability in direct compressible formulations containing novel excipients and different types of active ingredients. *Powder Technology*. 312: 222–236. <https://doi.org/10.1016/j.powtec.2017.02.019>
- Sipos, E., Oltean, A.R., Szabó, Z.I., Rédei, E.M., Nagy G.D., 2017. Application of SeDeM expert systems in preformulation studies of pediatric ibuprofen ODT tablets. *Acta Pharmaceutica*. 67(2): 237–246. <https://doi.org/10.1515/acph-2017-0017>
- Suñe, J.M., Roig Carreras, M., García, R.F., Montoya, E.G., Lozano, P.P., Aguilar, J.E., Carmona, M.M., Ticó-Grau, J.R., 2013. SeDeM Diagram: an expert system for preformation, characterization and optimization of tablets obtained by direct compression. *Formulation Tools for Pharmaceutical Development*. 109–135. <https://doi.org/10.1533/9781908818508.109>
- Suñe-Negre, J. M., Roig, M., Fuster, R., Hernández, C., Ruhí, R., García-Montoya, E., Pérez-Lozano, P., Miñarro, M., Ticó, J. R., 2014. New classification of directly compressible (DC) excipients in function of the SeDeM Diagram Expert System. *International Journal of Pharmaceutics*. 470: 15–27. <https://doi.org/10.1016/j.ijpharm.2014.04.068>
- Wan, S., Yang, R., Zhang, H., Li, X., Gu, M., Guan, T., Ren, J., Sun, H., Dai C., 2019. Application of the SeDeM Expert System in Studies for Direct Compression Suitability on Mixture of Rhodiola Extract and an Excipient. *An Official Journal of the American Association of Pharmaceutical Scientists*. 20(3): 105. <https://doi.org/10.1208/s12249-019-1320-4>
- Zhang, Y., Li, Y., Wu, F., Hong, Y., Shen, L., Lin, X., Feng, Y., 2021. Texture and surface feature-mediated striking improvements on multiple direct compaction properties of *Zingiberis Rhizoma* extracted powder by coprocessing with nano-silica. *International Journal of Pharmaceutics*. 603: 120703. <https://doi.org/10.1016/j.ijpharm.2021.120703>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.